

A peer-to-peer energy trading market embedded with residential shared energy storage units

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HIGHLIGHTS

- A peer-to-peer energy trading market mechanism with shared energy storage units is proposed.
- A non-cooperative game among peer-to-peer market participants is presented.
- The generalized Nash equilibrium problem among shared energy storage users is depicted.
- A game equilibrium solution based on KKT condition is proposed.
- Proposed mechanism improves the efficiency and ensures fairness.

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ABSTRACT

The high penetration of distributed energy resources with local renewable energy consumption facilitates the emergence of peer-to-peer (P2P) energy trading, where a peer can share excessive energy with local peers. P2P energy trading is expected to be a promising business model in the future transactive energy market. Influenced by the sharing economy principle, shared energy storage (SES) is investigated to reduce the costly initial investment and improve the utilization rate of storage devices. Given the context, this paper proposes an equilibrium model of a P2P energy trading market, considering the deployment of shared energy storage in the residential consumer-side. The market equilibrium problem consists of two interwoven games. Considering the benefit conflicts between energy sellers and buyers in P2P energy trading, the P2P transaction between peers can be described as a non-cooperative game. Considering the limited SES capacity resources, the competition among SES users can be viewed as a generalized Nash equilibrium problem. These two games in the P2P market interact with each other and can be summarized as a holistic equilibrium problem, in which the P2P transaction price is endogenously determined and storage capacity is allocated. The market equilibrium problem can be solved based on Karush-Kuhn-Tucker (KKT) optimality conditions and linearization techniques, finally cast as a mixed-integer linear program. Numerical simulation demonstrates that P2P energy trading benefits all participants and the introduction of shared energy storage can further reduce energy costs.

1. Introduction

1.1. Background

Energy is the foundation of human society and should be secured for sustainable and reliable development. Energy security is defined as the uninterrupted availability of energy sources at an affordable price [1], which has various dimensions, e.g., sources diversity, environmental

issues, energy cost, and sustainable energy consumption, etc [2]. Renewable energy generation and consumption are essential drivers to enhance energy security [3]. The modern power system is undergoing an unprecedented transformation towards renewable energy, with the ambition of reducing greenhouse gas emissions and improving energy security and sustainability. Distributed energy resources (DERs), such as solar photovoltaics, wind turbine, and energy storage devices, have been widely invested in distribution networks [4,5].

The massive installation of DERs in deregulated grid can provide

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Nomenclature			
Abbreviations			
DERs	Distributed energy resources	λ_{buy}^{bt}	The TLOU price of energy buyer b in period t
FiT	Feed-in Tariff	λ_{buy0}^t	Initial electricity price in TLOU in period t
GNEP	Generalized Nash equilibrium problem	k_t	Constant coefficient in TLOU
KKT	Karush-Kuhn-Tucker optimality conditions	p_s^t	The energy supply of seller s in period t
MILP	Mixed-integer linear program	D_b^t	The energy demand of buyer b in period t
P2P	Peer-to-peer	$\hat{I}_c$	Charging efficiency
SES	Shared energy storage	$\hat{I}_d$	Discharging efficiency
TLOU	Time-and-level-of-use	p^u	Maximal charging/ discharging power
Indices and sets		Q_b^0	Initial energy storage level of buyer b
B	Set of energy buyers	Q_m	Capacity of the shared energy storage units
S	Set of energy sellers	Variables	
b	The index of energy buyers	p_{sb}^t	P2P transaction amount between seller s and buyer b in period t
s	The index of energy sellers	ρ_{sb}^t	P2P transaction price between seller s and buyer b in period t
T	Set of time periods	p_{bg}^t	Energy from the grid of buyer b in period t
t	Index of time periods	p_{bc}^t	Charging power of buyer b in period t
Parameters		p_{bd}^t	Discharging power of buyer b in period t
Δt	Duration of period	Q_b	Capacity allocation of buyer b
λ_{sell}^t	Feed-in Tariff in period t	Q_b^t	Energy storage level of buyer b in period t
λ_{cost}^t	P2P transaction cost in period t		

benefits for both owners and the grid. On the one hand, the DERs owners can manage their energy consumption more effectively and receive revenue by selling excessive energy [6]. On the other hand, because of the massive scale of DERs, the distribution network can receive more flexibility by utilizing localized renewable energy generation [7]. Given the context, renewable energy consumption is an imperative task in the transactive energy market. A tariff-based scheme has been implemented to encourage DERs investment and improve energy consumption [8]. However, it has been criticized for lacking competition [9] and offering participants limited benefits [10]. The advancement in Information and Communication Technology and deregulations in distribution systems enable peer-to-peer (P2P) energy trading among local energy consumers and energy generators, which is expected to be a promising business model in the energy sharing market.

A peer-to-peer energy trading market consists of local energy buyers, sellers, and P2P service provider. A peer in the P2P market can directly buy/sell energy from/to other peers based on personal energy situation and energy price, without the intermediation by utility companies. All peers can participate in energy trading to gain financial benefit, thus creating a competitive energy trading market. From a peer's perspective, both energy buyers and sellers can achieve more flexible energy management, obtain financial benefits, and utilize excessive renewable energy. From the power system's perspective, local P2P energy trading can balance local energy supply and demand and reduce energy transmission losses, followed by improving power system reliability and postponing infrastructure investment [11]. P2P energy trading has attracted great attention in academia and industry. Many countries have conducted plenty of pilot projects to demonstrate the effectiveness of P2P energy trading in the transactive market [12], such as Piclo in the UK [13], Brooklyn Microgrid project in the USA [14], and Peer Energy Cloud in Germany [15].

Energy storage device, especially battery energy storage, is an essential component in the residential peer-to-peer energy trading market, enabling peers to schedule consumption profiles more flexibly, achieve peak-valley arbitrage and store excessive renewable energy [16,17]. Despite the economic and environmental benefits and declining installation cost [18], purchasing energy storage devices still requires costly investment and might encounter underutilization problems [19]

for a single household or residential community. Affected by the sharing economy principle, shared energy storage (SES) is an alternative way to reduce initial investment costs and improve utilization rate [20]. In the SES scheme, multiple energy consumers have access to use a shared energy storage device. From the perspective of investment, through the economy of scale and cost sharing, the installation cost is reduced compared with individual storage. From the perspective of operation, considering the temporal complementarity and various demand patterns among different users, SES can provide a larger capacity for users and increase utilization rate consequently. Considering a residential peer-to-peer energy trading market, where peers are embedded with shared energy storage, rather than individual energy storage, is an interesting and worthwhile topic.

1.2. Literature review

1.2.1. Peer-to-peer energy trading

P2P energy trading involves financial trading activities, which requires a proper business model design to stimulate peer's participation willingness and ensure the security of the system. Stakeholders' structures and energy pricing mechanism are two concerns when designing a business model.

The stakeholder's structure can describe the transaction relationships among participants and the division of transaction activities in a P2P market. In summary, the structure of stakeholders in the P2P market can be roughly classified into two types: decentralized market without an aggregator and centralized market with an aggregator.

- 1) **Decentralized market without an aggregator:** In a decentralized P2P market, peers can take full control of their trading strategies to maximize their interests or satisfy personal preferences [21], including transaction amount and price. Participants can negotiate or trade with each other directly without an aggregator. A bilateral contract of the transaction information between energy seller and buyer facilitates the P2P energy trading market, which is a simple type of over-the-counter (OTC) market in energy trading. In [22], a P2P market is proposed for real-time and forward bilateral contracts between energy suppliers and consumers. Participants have utility-

maximizing preferences with full substitutability, which ensures a stable result in the bilateral contracts network. In [23], the continuous double auction (CDA) method is used to match the transaction between energy sellers and buyers in a decentralized P2P market. Self-interested peers decide on their offers and bids based on specific strategies. In [24], a localized P2P energy trading market is implemented among plug-in hybrid electric vehicles (PHEVs). PHEVs determine their charging/discharging plan according to their driving plan, energy price, and current energy state, and then send energy trading requests to the nearest aggregator. Local aggregator acts as auctioneer and matches transactions between buyers and sellers.

- 2) **Centralized market with an aggregator:** In a centralized market, an aggregator can manage the energy consumption/supply of peers [16] or provide price signals for peers [6]. The P2P transactions are cleared through the aggregator and all energy tradings are under the supervision of the aggregator. In [25], a P2P energy trading model among households is proposed, where the total cost of all peers is minimized. A centralized aggregator collects the information from households and solves a multi-objective optimization problem to make sure all participants can benefit from the P2P trading, which addresses the unfair cost distribution problem. In [26], a centralized P2P trading environment is created for four types of households, based on whether they have rooftop PV and battery or not. The aggregator aims to minimize the total net energy cost of households and the optimization model is formulated as mixed-integer linear programming. In [27], a P2P energy trading platform is proposed for multiclass energy management and trading coordination between prosumers and the wholesale market. Energy is a heterogeneous product beyond financial characteristics and can be traded based on prosumers' personal preferences. A distributed price-directed optimization mechanism is developed to solve multiclass energy management.

The pricing mechanism is of great significance in P2P business model design [28]. Energy sellers are expected to receive a higher price in the P2P market than in the traditional power grid. Similarly, energy buyers are expected to purchase energy at a lower price from the P2P market than from the power grid. Such a pricing mechanism can provide financial incentives for local energy sellers and buyers. Otherwise, they don't have the incentive to participate in P2P energy trading. The existing pricing mechanisms include analytical methods [29,30], auction-based methods [23], game-theoretic methods [31], and blockchain-based methods [32]. We focus on the introduction of the auction-based method and game-theoretic method.

- 1) **Auction theory:** The auction theory is widely used in P2P transactive energy markets [33]. In [34], a multi-round double auction is established to complete the auction process. Peers participate in the hierarchical auction, including nodal, zonal, and distribution network levels to balance total local energy supply and demand. The clearing price is the average of offering price, which integrates with distributional locational marginal price, considering the power loss and voltage regulation. In [35], an iterative uniform-price auction mechanism is proposed to facilitate P2P energy trading in a community microgrid. Participants iteratively adjust their bids according to their personal demand/supply information and issued market information, aiming to maximize personal economic benefit. Such bidding strategy is different from the continuous double auction and leads to a Nash equilibrium among participants, which helps to find a fairer trading price. In [36], the Vickrey Clarke Groves Auction is used to motivate participants to bid at a rational price. The deployment of Bayesian Game Theory in bidding strategy stipulates efficient and cost-effective bidding for every energy buyer. Although the P2P transaction prices are derived from the auction process, their bidding strategies still involve game theory [35,36].

- 2) **Game theory:** Game theory is used in a competitive situation, where the result of one participant is associated with other participants' actions. The competitive characteristic requires participants should consider other participants' actions before taking their action. The P2P market is a competitive market, with energy sellers aiming to maximize energy revenues and energy buyers aiming to minimize energy costs. The game-theoretic method used in the P2P pricing mechanism can be divided into two parts: noncooperative game and cooperative game. For the **noncooperative game**, Nash equilibrium [37], generalized Nash equilibrium [38], and Stackelberg game [39] are widely applied in modeling P2P market. In [40], a P2P energy trading platform "ElecBay" is designed to facilitate P2P transactions among prosumers. A noncooperative game is used to model the bidding process of prosumers and a Nash equilibrium is obtained, in which no participant will change its strategy unilaterally. The bidding process converges to a generalized Nash equilibrium when prosumers consider their strategy's impact on energy trading prices [38]. In [41], a P2P energy sharing model considering the social attributes of prosumers is proposed based on the stochastic Stackelberg game. The leader is energy sharing provider with dynamic prices and the prosumers serve as followers, who adjust their energy consumption according to the price from the leader. An algorithm integrating interpolation and curve fitting, Markov decision process, and differential evolution algorithm is proposed to receive a Stackelberg equilibrium. For the **cooperative game**, in [42], the coalition graph game is used in P2P energy trading mechanism design to help prosumers form a coalition and choose the transaction quantity and price. The total payoff of the proposed P2P scheme is allocated among participants fairly under the Myerson value rule. A numerical study demonstrates that the P2P scheme can improve the financial benefits of prosumers while maintaining the constraint of the local voltage domain. In [43], a coalition formation game-based P2P energy trading scheme is developed to obtain a prosumer-centric solution. With the proposed algorithm, a prosumer can decide whether or not to put its battery into the P2P market and form a coalition.

However, there are still some research gaps in the existing peer-to-peer energy trading research works. In a centralized market with an aggregator, some researches consider P2P energy trading from a system perspective and ignore individual interest, which might violate personal interests [44]. Many existing pricing mechanisms in P2P energy trading can not reflect the true value of transactive energy under different supply-demand relationships [30]. Actually, a proper peer-to-peer energy trading business model should respect personal interests and ensure financial benefits, thus encouraging people to join the energy sharing market. Besides, the energy transaction price should reflect the scarcity of energy resources, e.g., a lower energy supply associated with a higher energy price and vice versa.

The successful implementation of the peer-to-peer energy trading mechanism still meets many risks and challenges. Peer-to-peer energy trading consists of both financial and physical activities. Financial activities should properly coordinate the conflicting interests of various participants while providing secure and reliable transaction information. The energy traded in financial activities should be transmitted via the physical network. As a consequence, the physical factors including the distribution network constraints, power transmission loss, unchecked power injection, etc., should be taken into consideration. Last but not least, the regulatory change is the premise of establishing a P2P energy trading market [45].

1.2.2. Shared energy storage

Shared energy storage has attracted more and more attention in the last decade [46]. The essence of shared energy storage is the separation of ownership and use rights of energy storage devices. At present, the mainstream structures of shared energy storage can be divided into two categories: 1) users own individual energy storage devices and share

their capacity use right or stored energy; 2) users connected to a central energy storage device and share the capacity or stored energy.

For the first category, the SES has been invested and installed by an individual user, who requires an incentive to share individual storage capacity or stored energy. In [47], individual storage device owners decide to share part of their capacity with shared facility controllers in response to the market price of capacity use right. The market price is determined by the auction process and encompasses incentive compatibility and individual rationality, which is testified to be an effective market mechanism. For the second category, the SES can be invested by independent investors or users and shared by multiple users. In [48], a cloud energy storage (CES) concept is proposed to utilize centralized energy storage devices and provide energy storage services for distributed users. Residential energy consumers or commercial users can rent the power and capacity of the CES to reduce personal energy costs.

For the successful implementation of shared energy storage, an economical and fair operation policy is needed to coordinate the complicated interactions of SES owners, users, and other stakeholders. Economical operation in this context means that shared energy storage should bring more financial benefits compared with individual storage. Fair operation refers to the fair distribution of financial benefits or storage capacity among SES users/owners. In [49], the impacts of individual and shared energy storage in the residential community are thoroughly compared and discussed, where shared energy storage brings more cost saving and higher utilization. In [50], the potential of deploying community storage is simulated based on the actual residential data. The results illustrate that the community energy storage decreases the needed storage amount, promotes renewable local consumption, and increases the Internal Rate of Return. However [49,50], mainly focus on evaluating the economic benefits of shared energy storage from a holistic perspective, without considering the fairness among SES users/owners. A shared energy storage service pricing scheme is proposed in [20], which ensures the service price of SES is fair among SES users. The proportional method [51], Shapley value method [52], nucleolus method based on coalitional game theory [53,54] are widely used to solve the fair benefit/cost allocation issue. A trading algorithm is proposed to ensure fairness in a generalized Nash game type capacity allocation problem among SES users [55].

Although many researches have investigated the deployment of shared energy storage, few studies consider the significance of shared energy storage in the P2P energy trading market, which requires more innovative researches in this field.

1.3. Contributions

To fill the aforementioned research gaps in peer-to-peer energy trading and shared energy storage, in this paper, we propose a novel P2P energy trading market and implement shared energy storage in the market. The main contributions of our work are twofold:

- 1) A novel peer-to-peer energy trading business model with shared energy storage installed in the energy consumer-side is proposed. In the P2P market, every participant is rational and self-interested, with the options of selling/buying energy to/from any other peers in the market and using SES units, which forms a competitive market and leads to a market equilibrium problem. Considering the bargain between energy sellers and buyers in the P2P market, the transaction between peers can be described as a non-cooperative game. Considering the limited shared energy storage resources of SES units, the competition between SES users is a generalized Nash equilibrium problem. These two games in the P2P market can be summarized as an equilibrium problem, in which the P2P transaction price is endogenously determined and SES capacity is allocated. However, the above two games are interwoven and the equilibrium problem is difficult to solve.

- 2) A solution method based on Karush-Kuhn-Tucker (KKT) optimality condition is applied to solve the market equilibrium problem. The market equilibrium is acquired based on the KKT conditions of all participants' optimal problems. After concatenating all KKT conditions together, the market equilibrium is cast as a mixed-integer linear program (MILP) with the complementarity conditions linearized through the Big-M method. The P2P transaction price and the amount and the SES capacity allocation are obtained from the market equilibrium. The P2P transaction price is determined by the supply and demand relationship and reflects the value of energy resources. Capacity allocation among users can ensure fairness in our definition.

The remainder of the paper is organized as follows. Section 2 introduces a representative system configuration of a P2P energy trading market embedded with residential shared energy storage. The mathematical model of the P2P market participants and the market equilibrium problem are presented in Section 3. Section 4 summarizes the solution technique of the equilibrium problem. Case studies are performed in Section 5. Finally, Section 6 concludes the paper.

2. System configuration

The most representative structure of the peer-to-peer energy trading market with shared energy storage units is shown in Fig. 1. In such a P2P market, a participant who has excessive energy and sells energy to other participants or the power grid is defined as a typical energy seller, e.g., a rooftop PV plant. A participant who consumes energy and buys energy from energy sellers or power grid is identified as a typical energy buyer, e.g., a community. Energy sellers and buyers in the P2P market can trade with each other freely and reach a bilaterally satisfactory transaction result. The solid arrows present the P2P transaction between typical energy sellers and energy buyers. All participants are connected with the local distribution network and the energy trading can be finished through physical power transmission lines.

The shared energy storage is implemented in our proposed P2P energy trading market, which is large-capacity energy storage and shared by multiple energy buyers. We assume the storage device has been invested and installed on the community side. In this paper, we just focus on the operational problem of shared energy storage. In our business model, the energy buyers can use the capacity of SES instead of equipping individual energy storage, which can save storage investment costs for energy buyers. After the capacity allocation, users can purchase more energy and charge SES when the electricity price is low and discharge SES when the price is high, thus reducing their energy costs.

The P2P market has an aggregator, who takes the responsibility of matching the P2P energy transaction and operating the SES.

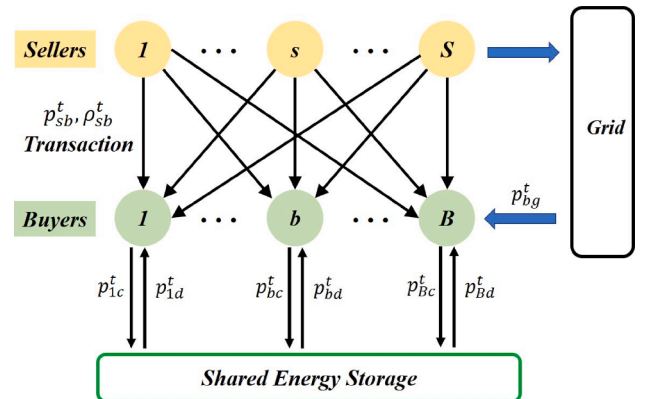


Fig. 1. P2P energy trading market with shared energy storage equipped in the community side.

In summary, the whole P2P market is composed of three types of stakeholders: energy sellers, energy buyers, and an aggregator. The detailed introduction of each stakeholder is presented as follows:

2.1. Energy sellers

Energy sellers consist of participants with renewable power generation, e.g., solar photovoltaics, wind turbine. These participants have surplus renewable energy and would like to share energy with other people in the P2P market to get a financial payoff. They can also choose to sell energy to the power grid in a traditional FiT scheme. Each energy seller, whose goal is to maximize personal profit, decides to participate in the P2P transaction market or trade with power utility according to the price signal.

Energy sellers' renewable energy supply is intermittent and volatile. However, to facilitate our P2P transaction, we assume each energy seller has been equipped with a storage device or other power electronic device to stabilize their energy supply in a certain time slot.

2.2. Energy buyers

Energy buyers consist of participants who consume energy, e.g., residential community, business building, shopping mall. We consider the residential community in this paper. Modeling the behavior of residential energy consumers requires both financial and emotional factors, integrating the economic and motivational psychology[56]. Besides, the ideal model is not always consistent with the real-world situation. So we choose the energy cost, which is a conspicuous and acceptable criterion to model consumer behavior. Each energy buyer, whose goal is to minimize the expenditure on energy purchase, can buy energy from the P2P market or power grid depending on the price signal. Besides, multiple buyers share the capacity use right of the SES, which provides the buyers with more flexible energy consumption management. They can use SES and adjust their charging/discharging strategy according to the electricity price on the premise of meeting the current energy deficit. To simplify the problem and show the operation of our P2P market, the initial energy demand of each energy buyer is fixed.

2.3. Aggregator

The aggregator is a third-party agency, who controls the operation of the P2P market. In the P2P market, energy sellers/buyers are rational and pursue to maximize personal interest. In each P2P transaction, a rational seller chooses to improve the transaction price to get more revenue; in contrast, a rational buyer chooses to lower the transaction price to pay less cost. Such a game induces a market equilibrium, where a mutually satisfactory price is obtained and no one will change his strategy unilaterally. P2P transaction price should be no less than FiT and no higher than TLOU, otherwise, energy sellers/buyers will quit the P2P market.

Since the energy buyers use the SES, the allocation of SES use right is also a game. In practice, the capacity resource of SES is limited. The capacity constraint is a global constraint shared by all SES users and competition exists when buyers' total capacity demand exceeds the SES capacity and. Such competition brings a new game, which will induce a capacity allocation equilibrium considering the fairness problem.

The aggregator collects all participants' supply and demand information and clears the P2P market. By solving the market equilibrium problem, it can provide P2P transaction price and amount between energy seller and buyer and capacity allocation among energy buyers. The aggregator can facilitate the P2P market and make sure participants benefit from the market.

3. Mathematic formulation

Mathematic formulations of the proposed P2P market involving

energy sellers, energy buyers, and market equilibrium are presented in the following subsection.

3.1. Seller's problem

In our model, the P2P market consists of energy buyers and sellers. Let S denote the set of energy sellers, with the typical energy seller $s \in S$. Let B denote the set of energy buyers, with the typical energy buyer $b \in B$. The operational problem is solved during multi-periods and T is the set of all periods. $t \in T$ is a typical single period with an equal duration of Δt .

For each rational energy seller $s \in S$, it maximizes its profit and the decision-making problem can be cast as

$$\max : \sum_{t \in T} \left(\sum_{b \in B} p_{sb}^t p_{sb}^t + \lambda_{sell}^t (P_s^t - \sum_{b \in B} p_{sb}^t) - \lambda_{cost}^t \sum_{b \in B} p_{sb}^t \right) \Delta t \quad (1)$$

$$\text{s.t.} \sum_{b \in B} p_{sb}^t \leq P_s^t : v_s^t, \forall t \in T \quad (2)$$

$$p_{sb}^t \geq 0 : \omega_{sb}^t, \forall b \in B, \forall t \in T \quad (3)$$

where p_{sb}^t and ρ_{sb}^t are decision variables. p_{sb}^t is the transaction amount between seller s and buyer b in period t . ρ_{sb}^t is transaction price associated with p_{sb}^t . However, the market-clearing price ρ_{sb}^t is out of control of seller s and buyer b but is determined by the market equilibrium problem of the game. Parameter P_s^t is the renewable energy for sale of seller s in period t . Parameter λ_{sell}^t is FiT when a seller sells its energy to the power grid. Parameter λ_{cost}^t is the P2P transaction cost, including network tariff for using the physical grid facilities and aggregator service fee.

The objective of each energy seller is to maximize its profit. In the objective function (1), the first term is the revenue of trading with energy buyers in the P2P market. The second term is the revenue of selling energy to the power grid in the FiT scheme. After participating in the P2P transaction, energy seller can still sell their surplus to the power grid to get more income. The third term is the transaction cost for P2P transactions. Constraint (2) stipulates that total P2P transaction amounts in period t should not exceed the seller's initial excessive energy. Constraint (3) is for the non-negativity condition for the decision variable p_{sb}^t . v_s^t and ω_{sb}^t after the colon are dual variables of the constraints.

3.2. Buyer's problem

For each rational energy buyer $b \in B$, it minimizes energy cost and the decision-making problem can be cast as

$$\min : \sum_{t \in T} \left(\sum_{s \in S} p_{sb}^t p_{sb}^t + \lambda_{buy}^{bt} p_{bg}^t \right) \Delta t \quad (4)$$

$$\text{s.t.} \sum_{s \in S} p_{sb}^t + p_{bg}^t + p_{bd}^t = D_b^t + p_{bc}^t : \gamma_b^t, \forall t \in T \quad (5)$$

$$p_{bg}^t \geq 0 : \omega_{bg}^t, \forall t \in T \quad (6)$$

$$\lambda_{buy}^{bt} = \lambda_{buy0}^t + k_t p_{bg}^t, \quad \forall t \in T \quad (7)$$

$$p_{bc}^t \leq p^u, p_{bd}^t \leq p^u, \quad \forall t \in T \quad (8)$$

$$0 \leq p_{bc}^t \perp p_{bd}^t \geq 0, \quad \forall t \in T \quad (9)$$

$$Q_b^t = Q_b^{t-1} + (p_{bc}^t \eta_c - p_{bd}^t / \eta_d) \Delta t, \quad \forall t \in T \quad (10)$$

$$0 \leq Q_b^t \leq Q_b, \quad \forall t \in T \quad (11)$$

$$\sum_{b \in B} Q_b \leq Q_m : \tau_b, \quad \forall t \in T \quad (12)$$

where p_{sb}^t , p_{sb}^t , p_{bg}^t , p_{bc}^t , p_{bd}^t and Q_b are decision variables. p_{sb}^t and p_{sb}^t have been introduced in the seller's decision-making problem. p_{bg}^t is the energy from the power grid in period t . p_{bc}^t and p_{bd}^t are buyer's charging/discharging strategy in period t . Variable Q_b is the capacity allocated to energy buyer b . Parameter λ_{buy}^{bt} is the time-and-level-of-use price for buyer b purchase electricity from the power grid in period t , which is composed of an initial price part λ_{buy0}^t and a level-wise part $k_t p_{bg}^t$, as shown in constraint (7). k_t is a constant in period t . η_c/η_d are charging/discharging efficiency of energy storage device. Q_b^t is the storage level of buyer b in period t . Q_m is the total capacity of SES.

The objective of each energy buyer is to minimize its energy cost. In the objective function (4), the first term is the total cost of trading with energy sellers in the P2P market. The second term is the cost of purchasing energy from the power grid. Constraint (5) ensures the power balance in period t , where the total energy demand including fixed demand D_b^t and charging demand p_{bc}^t should be met by the energy from the P2P market, power grid, and discharging SES. Constraint (6) imposes a non-negativity condition on the decision variable p_{bg}^t . Constraint (8) and (9) provide an upper bound and lower bound for the charging and discharging power. And constraint (9) indicates SES users should not charge and discharge at the same time because of the energy loss during the charging and discharging process. Constraint (10) updates the SES energy storage level, which is a multi-period dynamic constraint. The storage level Q_b^t is consistent with the energy amount that buyer b stores in SES. Constraint (11) is the feasible range of the variable Q_b^t . Constraint (12) is a commonly shared constraint, which means the total capacity used by energy buyers should not exceed the maximum available capacity of SES.

To facilitate the problem description and solution, we make some reformulations. A rational SES user will not choose to charge and discharge simultaneously and thus the complementary slackness constraint (9) binds. The constraint (8) and (9) can be transformed as:

$$\begin{aligned} 0 \leq p_{bc}^t \leq p^u \\ 0 \leq p_{bd}^t \leq p^u \end{aligned} \quad (13)$$

The dynamic characteristic of energy storage level Q_b^t can be reformulated as:

$$Q_b^t = Q_b^0 + \sum_{i=1}^t (p_{bc}^i \eta_c - p_{bd}^i / \eta_d) \Delta t \quad (14)$$

where Q_b^0 is the initial energy storage level.

In summary, energy buyer's objective problem (4)-(12) can be transformed as follows:

$$\min : \sum_{t=1}^T \left(\sum_{s \in S} p_{sb}^t p_{sb}^t + \lambda_{buy}^{bt} p_{bg}^t \right) \Delta t \quad (15)$$

$$\text{s.t.} \sum_{s \in S} p_{sb}^t + p_{bg}^t + p_{bd}^t = d_b^t + p_{bc}^t : \gamma_b^t, \forall t \in T \quad (16)$$

$$p_{bg}^t \geq 0 : \omega_{bg}^t, \forall t \in T \quad (17)$$

$$\lambda_{buy}^{bt} = \lambda_{buy0}^t + k_t p_{bg}^t, \forall t \in T \quad (18)$$

$$p_{bc}^t \geq 0 : \delta_{bc0}^t, \forall t \in T \quad (19)$$

$$p_{bc}^t \leq p^u : \delta_{bc1}^t, \forall t \in T \quad (20)$$

$$p_{bd}^t \geq 0 : \delta_{bd0}^t, \forall t \in T \quad (21)$$

$$p_{bd}^t \leq p^u : \delta_{bd1}^t, \forall t \in T \quad (22)$$

$$Q_b^0 + \sum_{i=1}^t (p_{bc}^i \eta_c - p_{bd}^i / \eta_d) \Delta t \geq 0 : \theta_{b0}^t, \forall t \in T \quad (23)$$

$$Q_b^0 + \sum_{i=1}^t (p_{bc}^i \eta_c - p_{bd}^i / \eta_d) \Delta t \leq Q_b : \theta_{b1}^t, \forall t \in T \quad (24)$$

$$\sum_{b \in B} Q_b \leq Q_m : \tau_b, \forall t \in T \quad (25)$$

The objective function (15) is the cost minimization problem of energy buyer b . Constraint (16) is the energy balance equation. Constraint (17) is for the non-negativity condition for p_{bg}^t . Constraint (18) is the time-and-level-of-use price for buying electricity from the power grid. Constraint (19)-(22) are the feasible ranges of charging and discharging rate. Constraint (23) and (24) are the transformation of constraints (11), considering the dynamic constraint (14). Constraint (25) is still the commonly shared capacity constraint. γ_b^t , ω_{bg}^t , δ_{bc0}^t , δ_{bc1}^t , δ_{bd0}^t , δ_{bd1}^t , θ_{b0}^t , θ_{b1}^t and τ_b after the colon are dual variables of the constraints.

3.3. Peer-to-peer market equilibrium problem

In the P2P market, all participants want to maximize their interests. However, there might be interests conflicts among the participants. The P2P market game model can be cast as an equilibrium model:

$$\begin{cases} \text{Seller's Problem (1) - (3),} & s \in S \\ \text{Buyer's Problem (15)-(25),} & b \in B \end{cases} \quad (26)$$

Equilibrium problem (26) entails solving all participants' problems at the same time. The problem is solved by the P2P market aggregator, and the solution method is proposed in the next section to solve the P2P market equilibrium problem. There are two games in the equilibrium model, a non-cooperative game and a generalized Nash equilibrium problem (GNEP), which are depicted in Fig. 2.

For the first game, energy sellers want to maximize energy revenues and will increase the P2P transaction price p_{sb}^t . In contrast, energy buyers aim to minimize energy costs and will lower the P2P transaction price p_{sb}^t . Considering the bargain between energy sellers and buyers in the P2P market, the transaction between peers can be described as a non-cooperative game. The first game equilibrium exists when considering energy sellers and buyers' optimal problems simultaneously. The P2P transaction price p_{sb}^t is an endogenous variable, which is not determined by a single participant unilaterally, but obtained from the market equilibria according to the supply and demand relationship. So the P2P transaction price can reflect the value of energy resources.

For the second game, the SES capacity is a global shared but limited resource. Competition arises when the capacity demand exceeds the available capacity of SES, which forms a generalized Nash equilibrium problem. The second game equilibrium exists when considering the capacity allocation among energy buyers. Constraint (25) imposes global storage capacity constraint for SES users, followed by a Lagrange dual multiplier τ_b . The dual variable has a clear economic interpretation: since limited storage capacity is a scarce resource, τ_b can be viewed as the shadow prices paid by buyers for using SES capacity. Actually, when capacity is adequate, the inequality constraint is not binding and the dual variable is zero; the dual variable is positive only if capacity demand exceeds the available capacity of SES, indicated by a binding inequality constraint [57].

The GENP is cast as a normalized Nash equilibrium [58] when the Lagrange dual multiplier τ_b incorporates the following cone constraint:

$$\tau_b = \beta_b, \tau_0, \beta_b > 0, b \in B \quad (27)$$

where β_b is assumed to be directly proportional to the average daily energy demand of buyer b . The assumption is reasonable because consumers with more energy demand usually would like to install a larger energy storage device, reflecting energy storage values are different among various consumers. In our paper, we refer to fairness as the fair capacity allocation based on the storage value of SES for users, which is indicated by the proportional Lagrange dual multiplier associated with

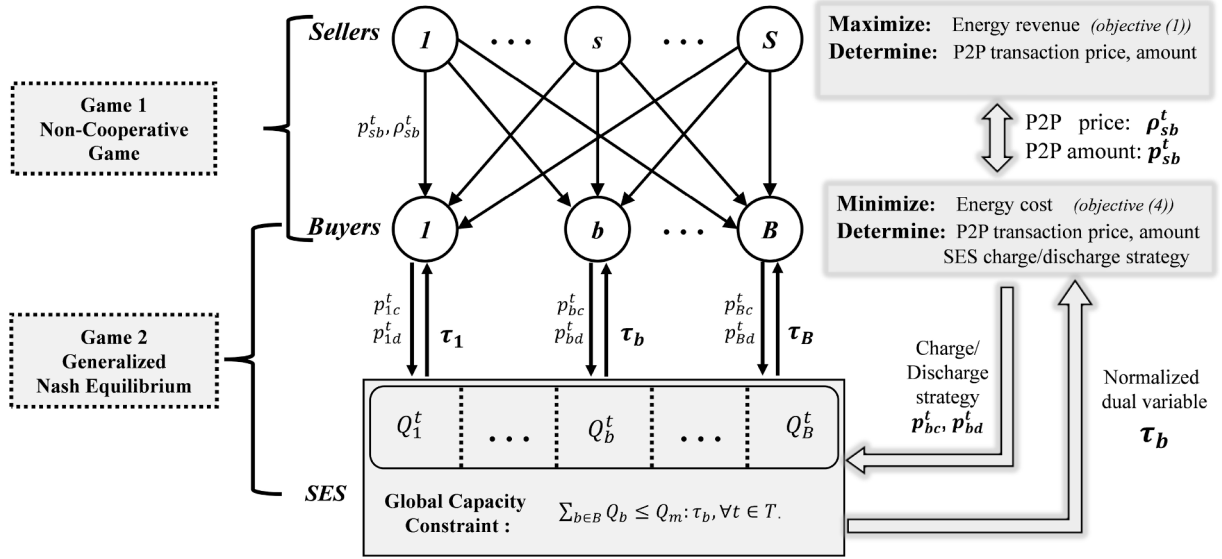


Fig. 2. Market equilibrium consists of two games: a non-cooperative game and a generalized Nash equilibrium problem.

global storage capacity constraint. The different Lagrange dual multipliers are implemented in the following solution method to conduct capacity allocation.

4. Solution method

In equilibrium problem (26), energy seller s and buyer b have consistent decision variables p_{sb}^t and p_{sb}^t because of the P2P transaction. However, the product term $\rho_{sb}^t p_{sb}^t$ in decision-making problem (1) and (15) makes the problem nonconvex. Besides, the transaction price ρ_{sb}^t is an endogenous variable beyond the control of single energy seller s or buyer b , which is determined from the market equilibrium condition. So the equilibrium problem (26) is not a Nash-type game. To explain how the market equilibrium arises and how the P2P transaction price is obtained, we formulate the KKT conditions of the equilibrium problem (26) in this section.

The detailed formulations are as follows:

4.1. KKT conditions of the market equilibrium problem

Since transaction price ρ_{sb}^t is an endogenous variable and derived from market equilibrium, we regard it as a parameter while obtaining the KKT conditions of energy sellers and buyers. If we consider the optimization problem (1) and (15) from this perspective, the initial objective function is a linear program and the KKT condition of the linear program is necessary and sufficient. After getting the KKT conditions of all participants, we concatenate all KKT conditions and get an equilibrium problem of KKT conditions.

The KKT condition of energy seller s obtained from seller's problem (1)-(3) is as follows:

$$(-\rho_{sb}^t + \lambda_{sell}^t + \lambda_{cost}^t)\Delta t + v_s^t - \omega_{sb}^t = 0, \quad \forall b \in B, \forall t \in T \quad (28)$$

$$0 \leq (p_s^t - \sum_{b \in B} p_{sb}^t) \perp v_s^t \geq 0, \quad \forall t \in T \quad (29)$$

$$0 \leq p_{sb}^t \perp \omega_{sb}^t \geq 0, \quad \forall b \in B, \quad \forall t \in T \quad (30)$$

The expression $0 \leq a \perp b \geq 0$ is a complementary slackness constraint, which means both a and b are non-negative and at least one of them is 0. In vector notation, $a \perp b$ means $a^T b = 0$.

The KKT condition of energy buyer b obtained from buyer's problem (15)-(25) is as follows:

$$\sum_{s \in S} p_{sb}^t + p_{bg}^t + p_{bd}^t = D_b^t + p_{bc}^t, \quad \forall t \in T \quad (31)$$

$$\rho_{sb}^t \Delta t + \gamma_b^t = 0, \quad \forall s \in S, \quad \forall t \in T \quad (32)$$

$$(\lambda_{buy0}^t + 2k_t p_{bg}^t) \Delta t + \gamma_b^t - \omega_{bg}^t = 0, \quad \forall t \in T \quad (33)$$

$$0 \leq p_{bg}^t \perp \omega_{bg}^t \geq 0, \quad \forall t \in T \quad (34)$$

$$-\gamma_b^t - \delta_{bc0}^t + \delta_{bc1}^t + \left(-\sum_{t'=t}^T \theta_{b0}^{t'} + \sum_{t'=t}^T \theta_{b1}^{t'} \right) \eta_c \Delta t = 0, \quad \forall t \in T \quad (35)$$

$$0 \leq p_{bc}^t \perp \delta_{bc0}^t \geq 0, \quad \forall t \in T \quad (36)$$

$$0 \leq (p^t - p_{bc}^t) \perp \delta_{bc1}^t \geq 0, \quad \forall t \in T \quad (37)$$

$$\gamma_b^t - \delta_{bd0}^t + \delta_{bd1}^t + \left(\sum_{t'=t}^T \theta_{b0}^{t'} - \sum_{t'=t}^T \theta_{b1}^{t'} \right) \Delta t / \eta_d = 0, \quad \forall t \in T \quad (38)$$

$$0 \leq p_{bd}^t \perp \delta_{bd0}^t \geq 0, \quad \forall t \in T \quad (39)$$

$$0 \leq (p^t - p_{bd}^t) \perp \delta_{bd1}^t \geq 0, \quad \forall t \in T \quad (40)$$

$$0 \leq \left[Q_b^0 + \sum_{t'=1}^t (p_{bc}^{t'} \eta_c - p_{bd}^{t'} / \eta_d) \Delta t \right] \perp \theta_{b0}^{t'} \geq 0, \quad \forall t \in T \quad (41)$$

$$0 \leq Q_b - (Q_b^0 + \sum_{t'=1}^t (p_{bc}^{t'} \eta_c - p_{bd}^{t'} / \eta_d) \Delta t) \perp \theta_{b1}^{t'} \geq 0, \quad \forall t \in T \quad (42)$$

$$-\sum_{t'=1}^T \theta_{b1}^{t'} + \tau_b = 0, \quad \forall b \in B \quad (43)$$

$$0 \leq Q_m - \sum_{b \in B} Q_b \perp \tau_b \geq 0, \quad \forall b \in B \quad (44)$$

Finally, the KKT condition of equilibrium problem (26) can be summarized as:

$$\begin{cases} \text{Seller's KKT Condition (28) - (30),} & s \in S \\ \text{Buyer's KKT Condition (31) - (44),} & b \in B \end{cases} \quad (45)$$

KKT system (45) is composed of equations and inequality

constraints. The P2P transaction price variable ρ_{sb}^t is endogenously determined and the SES capacity is allocated by solving equilibrium (45). From the energy sellers' side, according to equation (28), ρ_{sb}^t can be calculated as follows:

$$\rho_{sb}^t = \lambda_{sell}^t + \lambda_{cost}^t + (v_s^t - w_{sb}^t)/\Delta t \quad (46)$$

From the energy buyers' side, according to equation (32), ρ_{sb}^t can be calculated as follows:

$$\rho_{sb}^t = -\gamma_b^t/\Delta t \quad (47)$$

The price in (46) and (47) are the same in the equilibrium problem (45), and thus the transaction price ρ_{sb}^t mainly depends on the parameters and some dual variables. This P2P pricing mechanism is beyond the control of a single participant, and truly reflects the demand and supply relationship and the true value of energy, which is also fair to both sides. In addition, in our model, ρ_{sb}^t must be no less than the $\lambda_{sell}^t + \lambda_{cost}^t$, otherwise, sellers will quit the P2P market. Similarly, ρ_{sb}^t must be no more than TLOU, otherwise, buyers will quit the P2P market.

4.2. Big-M method

The equilibrium problem (45) is a nonlinear problem and intrinsically hard due to the introduction of complementary slackness constraints. To cope with the complementary slackness constraints and retrieve the market equilibria, this section introduces a big-M method proposed in [59].

The big-M method is used for linearization. The standard compact linear complementary slackness constraints can be written as follows:

$$0 \leq Px + q \perp y \geq 0 \quad (48)$$

where x and y are primal and dual variables, respectively; P is a constant square matrix, b is a constant vector. If the primal constraint is not binding, its dual variable is zero; similarly, if the dual variable is positive, the primal constraint is binding. Introducing an auxiliary binary vector z , (48) can be reformulated as follows:

$$\begin{aligned} 0 &\leq Px + q \leq M(1 - z) \\ 0 &\leq y \leq Mz \end{aligned} \quad (49)$$

where M is a large enough constant. On the one hand, it should be large to cover the maximum points; on the other hand, a smaller value contributes to the convergence and computation process. However, constraint (49) lacks an optimization problem. A mixed-integer linear program subject to (49) is proposed as follows:

$$\min : \quad z(Px + q) + (1 - z)y \quad (50)$$

If the problem is feasible, the optimal value of (50) must be 0; otherwise, the optimal value is strictly positive. The product term $z_i P_{ij} x_j$ is composed of a binary variable and primal variable, which can be linearized as a linear term $P_{ij} z_{ij}^1$ by an additional variable z_{ij}^1 , subject to:

$$0 \leq z_{ij}^1 \leq M, \quad 0 \leq x_j - z_{ij}^1 \leq M(1 - z_i), \quad \forall i, j \quad (51)$$

Constraint (51) is denoted by $\text{Cons-Lin}(x; z)$. Similarly, the product term $z_i y_i$ can be reformulated by a scalar variable z_i^2 , subject to:

$$0 \leq z_i^2 \leq M z_i, \quad 0 \leq y_i - z_i^2 \leq M(1 - z_i), \quad \forall i \quad (52)$$

Constraint (52) is denoted by $\text{Cons-Lin}(y; z)$.

The final MILP model can be written as follows:

$$\min_{x, y, z, z^1, z^2} z^T q + y + \sum_{ij} P_{ij} h_{ij}^1 - \sum_i h_i^2 \text{ s.t. } (49), (x, y) \in \Gamma, \quad z \text{ is binary } \text{Cons-Lin}(x, z), \text{Cons-Lin}(y, z) \quad (53)$$

where Γ represents equality constraints in (45).

5. Case study

5.1. Test system configuration

To numerically testify the performance of the proposed P2P model, we collect actual load data from the Irish residential community during the year 2009. Four energy buyers' load profiles in a single day are shown in Fig. 3a, with a time granularity of 1 h. Two energy sellers' PV generation data curve in a single day is shown in Fig. 3b, with the assumption that the supply is stable in 1 h. Fig. 3c is the total energy supply and demand. In the time-and-level-of-use pricing scheme, the initial price signal TLOU0 and energy buyers' electricity prices without P2P energy trading are shown in Fig. 4. All data sets have been processed and scaled to validate our results. In practice, the load profiles, renewable energy generation, and electricity price can be obtained from the historical data combined with big data techniques.

For the time-and-level-of-use electricity price, the constant coefficient $k_t = 0.001$. The feed-in tariff is proportionate to the initial price signal, e.g., 0.7. Transaction cost $\lambda_{cost}^t = 0.005$. In price setting, the sellers are forbidden to buy from the grid and sell to the P2P market simultaneously and the time-and-level-of-use price is higher than the feed-in tariff, otherwise, an arbitrage opportunity arises. As for the shared energy storage, the capacity of the shared energy storage unit is $Q_m = 100\text{kWh}$, with charging/discharging efficiency $\eta_c = \eta_d = 0.95$. All experiments are carried out on a 64-bit laptop with Intel Core i7-7500U CPU with 2.7 GHz and 8 GB RAM. The equilibrium problem is coded in MATLAB with the YALMIP interface. Commercial solver MOSEK 9.1 is used to solve the MILP.

To demonstrate the advantage of the proposed P2P model embedded with residential shared energy storage, five case study scenarios are defined. Case 1 is the benchmark without any P2P energy trading and energy storage. In case 2, only P2P energy trading is taken into account. Case 3 is the model proposed in this paper, both P2P energy trading and shared energy storage are introduced. To compare the advantage of SES with individual energy storage, case 4 considers the individual storage installed by each buyer. Case 5 is an additional case study, using the method proposed in [49] to show the potential of shared energy storage based on our parameter setting. Case details are listed in Table 1.

5.2. Results and discussion

5.2.1. Benefit of P2P market embedded with SES

Based on Case 1 and Case 3, we illustrate the benefits of our proposed P2P market embedded with residential SES. The computation time for solving the MILP (53) is less than 10 s and the optimality gap is less than $\varepsilon = 10^{-6}$.

Table 2 summarizes the total cost of buyers during one day. Without the P2P market, buyers can only purchase electricity from the grid at TLOU price. With the P2P market, besides the grid, buyers can also purchase electricity from the P2P market at a relatively lower P2P price. The implementation of shared energy storage provides more flexible energy management by charging/discharging. The simulation results show that all energy buyers obtain cost savings, and the total cost reduction is up to 18.27%. Notably, buyer 4 enjoys the biggest cost decline among the buyers, which will be discussed in the next subsection.

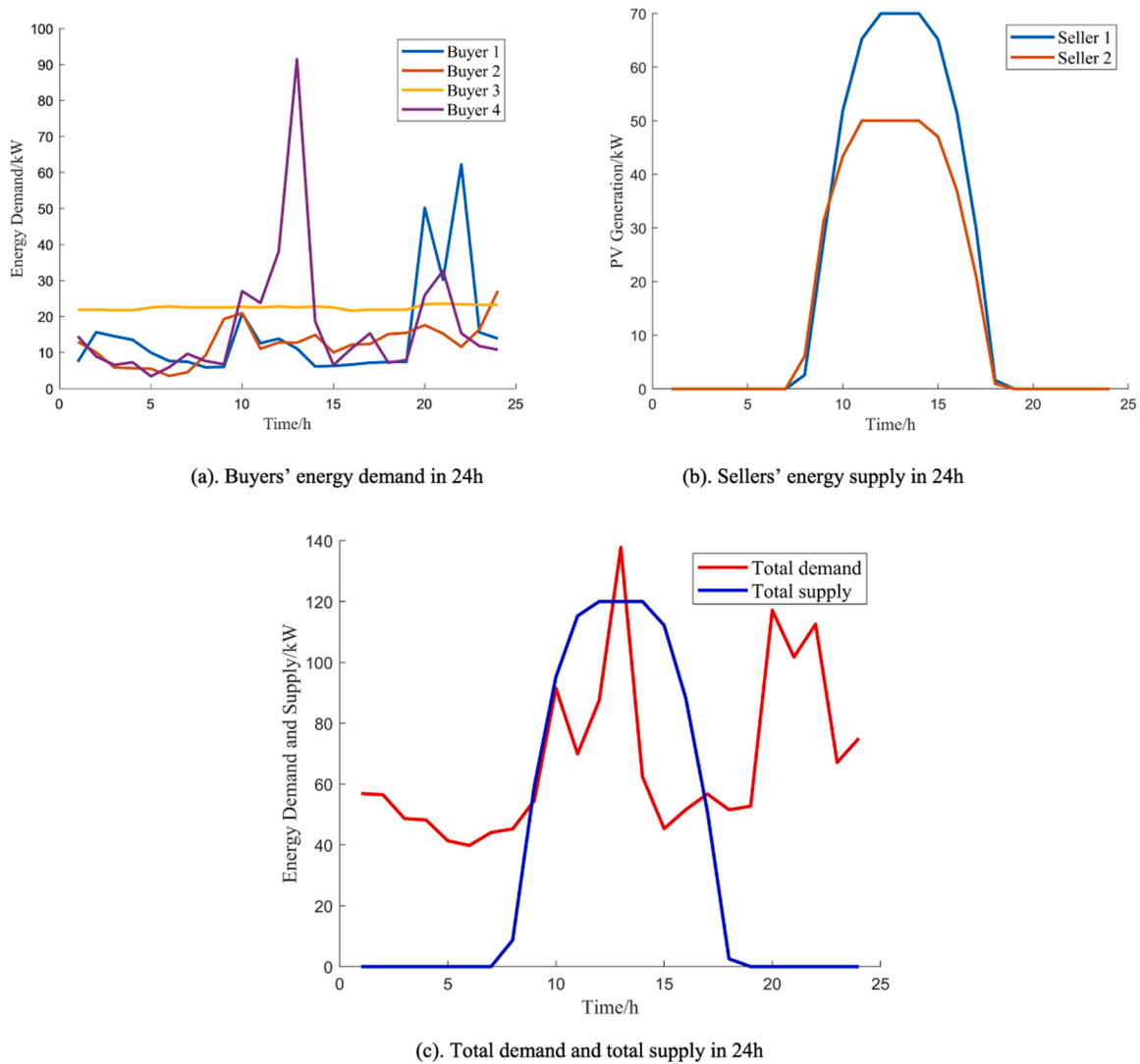


Fig. 3. Supply and demand information.

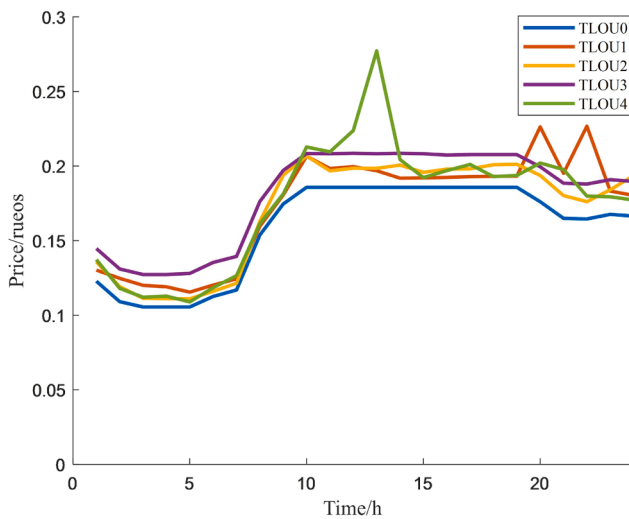


Fig. 4. Real-time electricity price in 24 h.

Table 1

Summary of 5 study case scenarios.

Scenario	p2p	SES	individual
case 1	×	×	×
case 2	✓	×	×
case 3	✓	✓	×
case 4	✓	×	✓
case 5	×	✓	×

Table 2

Summary of the cost of buyers without and with P2P market

buyer	case 1	case 3	benefit	percentage of benefit
B1	67.65	59.25	8.40	12.41%
B2	55.03	46.01	9.02	16.40%
B3	97.84	84.41	13.42	13.72%
B4	85.67	60.57	25.10	29.30%
total	306.19	250.25	55.95	18.27%

Table 3 summarizes the total revenue of energy sellers during one day. Without the P2P market, they can only feed their surplus energy back to the grid. With the P2P market, they can both feed into the grid and share with other P2P market participants. It can be seen that the

Table 3

Summary of the revenue of sellers without and with P2P market

SELLER	case 1	case 3	benefit	percentage of benefit
S1	65.47	67.01	1.54	2.35%
S2	49.90	51.84	1.94	3.88%
TOTAL	115.38	118.85	3.47	3.01%

revenues of energy sellers are improved. The total revenue has been increased by 3.01%. There are differences between the cost of buyers and the revenue of sellers, which can be explained as follows. The PV generations of energy sellers are concentrated between 10 a.m. and 4p.m. as shown in Fig. 3c. However, during the generation peak time, total supply exceeds total demand at most time periods, except 1 p.m. Considering the supply–demand ratio, sellers will receive a lower P2P transaction price and limited benefit. Conversely, buyers will enjoy a higher price reduction compared with buying electricity from the grid at TLOU price. So the benefits of energy buyers and sellers are different.

In conclusion, according to Tables 2 and 3, all participants can benefit from the P2P market embedded with SES in our model.

5.2.2. Peer-to-peer market equilibrium

To further show the results of P2P transactions, we pick up 4 typical periods to analyze the P2P market equilibrium in Case 3, as shown in Table 4. The transaction pair is (p_{sb}^t, p_{st}^t) , indicating the P2P transaction price and amount between seller s and buyer b during period t . The supply is the PV generation, while the demand is the initial fixed demand plus/minus charging/discharging power of shared energy storage. TLOU0 is the base price faced by buyers when purchasing electricity from the grid. FiT is the Feed-in Tariff offered by the grid. In our model, the P2P transaction price and amount are obtained from solving the market equilibrium problem. Table 4 provides a direct insight into how market equilibrium arises.

When $t = 8$, the PV generation is small, so the total energy demand among energy buyers exceeds the total energy supply. In such a circumstance, all buyers will compete for the limited energy resource. Buyers offering higher transaction prices will acquire the limited energy, e.g., B3 with the highest price 0.1814. Actually, the price 0.1814 is also the marginal cost when B3 buys electricity from the grid to satisfy the unmet needs after the P2P transaction. The same analysis works at $t = 18$ time period.

When $t = 15$, the total PV generations exceed the total energy demand of buyers, which means the renewable energy generations can not only satisfy all P2P transaction demand but also have to sell excess energy to grid. Given the context, buyers will trade with sellers who require a lower price. And sellers have the motivation to decrease their

price in exchange for competitiveness in P2P transactions. A conspicuous feature is that each buyer purchases energy from different sellers with the same price 0.1350. The reason is that if a seller improves the P2P transaction price, buyers will not buy energy from him and turn to other sellers with lower prices, causing a revenue drop to the aforementioned seller. Besides, the same price is the FiT plus the transaction cost λ^{cost} . FiT can be seen as the marginal revenue of selling energy to the grid, which is identical to all sellers. The same parameter setting for all sellers accounts for the same P2P transaction price.

Consider the final scenario $t = 13$, when total demand is equal to total supply. Actually, the scenario is the same as previously mentioned $t = 15$ scenario. Besides taking up the most P2P transaction amount, B4 also discharges 12.82 kWh from shared energy storage to satisfy the sharply increasing demand. The discharging behavior means the buyers can use their stored energy in SES according to personal willingness. In other words, when buyers discharge more energy from SES, the total demand is less than the total supply, reduced to scenario $t = 15$. B4 can meet most energy demand from the P2P market at a relatively lower price, explaining why B4 enjoys the biggest cost decline.

5.2.3. Operation of shared energy storage

The simulation results of Case 2 and Case 3 are summarized in Table 5. It can be seen that, compared with Case 2 without energy storage, the total cost saving for entire buyers embedded with shared energy storage is 5.46%. The cost saving is an expected result because shared energy storage enables users to manage their energy consumption more flexibly and achieve peak-valley arbitrage. Total P2P amount refers to the summation of P2P transactions during 24 h. From the comparison of the results, the deployment of shared energy storage devices improves the total P2P transaction amount, indicating SES can further promote local renewable consumption and energy trading activities.

The capacity allocation among buyers is (39.04, 22.22, 0, 38.74) kWh,

Table 5

Summary of the simulation results for P2P model without SES and with SES

buyer	Case 2		Case 3			
	Cost	Total P2P amount	Capacity allocation	Cost	Cost savings	Total P2P amount
B1	62.98	84.73	39.04	59.25	5.92%	98.74
B2	48.22	120.29	22.22	46.01	4.58%	134.37
B3	85.57	207.13	0	84.41	1.35%	211.77
B4	67.94	232.80	38.74	60.57	10.85%	263.58
total	264.71	644.96	100.00	250.25	5.46%	708.46

Table 4

P2P transaction information in 4 typical time periods.

t = 8					t = 13				
TLOU0 = 0.1538, FiT = 0.1076, Demand > Supply					TLOU = 0.1857, FiT = 0.1300, Demand = Supply				
	S1	S2	Total			S1	S2	Total	
Supply/Demand	2.59	6.10	8.69		Supply/Demand	70.00	50.00	120.00	
B1	0	(0.1538, 0.00)	(0.1538, 0.00)	0	B1	8.37	(0.1350, 8.37)	(0.1350, 0.00)	8.37
B2	0	(0.1496, 0.00)	(0.1496, 0.00)	0	B2	10.52	(0.1350, 0.00)	(0.1350, 10.52)	10.52
B3	22.50	(0.1814, 2.59)	(0.1814, 6.10)	8.69	B3	22.50	(0.1350, 22.50)	(0.1350, 0.00)	22.50
B4	0	(0.1361, 0.00)	(0.1361, 0.00)	0	B4	78.61	(0.1350, 39.13)	(0.1350, 39.48)	78.61
Total	22.50	2.59	6.10		Total	120.00	70.00	50.00	
t = 15					t = 18				
TLOU = 0.1857, FiT = 0.1300, Demand < Supply					TLOU = 0.1857, FiT = 0.1300, Demand > Supply				
	S1	S2	Total			S1	S2	Total	
Supply/Demand	65.17	47.00	112.1700		Supply/Demand	1.61	0.95	2.56	
B1	14.48	(0.1350, 0.00)	(0.1350, 14.48)	14.48	B1	7.36	(0.2005, 0.00)	(0.2005, 0.00)	0
B2	10.02	(0.1350, 0.00)	(0.1350, 10.02)	10.02	B2	5.77	(0.1973, 0.00)	(0.1973, 0.00)	0
B3	22.50	(0.1350, 0.00)	(0.1350, 22.50)	22.50	B3	21.90	(0.2244, 1.61)	(0.2244, 0.95)	2.56
B4	19.34	(0.1350, 19.34)	(0.1350, 0.00)	19.34	B4	7.18	(0.2001, 0.00)	(0.2001, 0.00)	0
Total	66.34	19.34	47.00		Total	42.21	1.61	0.95	

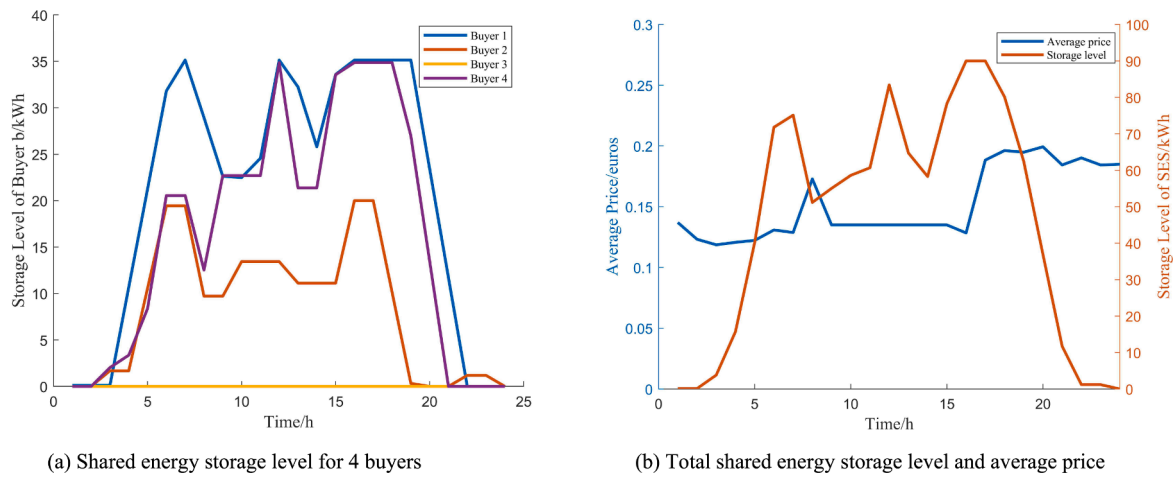


Fig. 5. One day operation of shared energy storage.

Table 6

Summary of the simulation results for shared and individual energy storage.

buyer	CASE 2	CASE 3			CASE 4		
	Cost	Capacity	Cost	Cost savings	Capacity	Cost	Cost savings
B1	62.98	39.04	59.25	5.92%	22.25	60.77	3.51%
B2	48.22	22.22	46.01	4.58%	18.70	46.51	3.53%
B3	85.57	0	84.41	1.35%	33.42	82.94	3.07%
B4	67.94	38.74	60.57	10.85%	25.63	62.83	7.52%
total	264.71	100.00	250.25	5.46%	100.00	253.05	4.40%

which is a generalized Nash equilibrium corresponding to our proposed model. Since our model has two interwoven games, it is difficult to analyze the capacity allocation. We conduct Case 5 by adopting the SES model proposed in [49] to validate the allocation result. It is worth mentioning that Case 5 does not have P2P energy trading and the optimized objective function is minimizing all users' energy costs from a holistic perspective. Despite the above differences, Case 5 can still provide a referable result. The capacity allocation in Case 5 is (26.63, 0.29, 0, 73.09) kWh, where B3 has no capacity, B1 and B4 take up most of the capacity. The result of Case 5 can validate the allocation result in Case 3 to some extent.

Fig. 5a shows the one-day energy storage level for each buyer. Fig. 5b is a double vertical axis graph and shows the operation of shared energy storage devices from a holistic perspective. The left coordinate axis corresponds to the average energy price for 4 buyers, integrating TLOU price and P2P transaction price, which can provide an intuitive impression on the price signal, despite not being that accurate. The right coordinate axis corresponds to the total energy storage level of the shared energy storage system. During 1 a.m. and 7 a.m., the electricity price is relatively low and users start to charge. During 8 a.m. and 4 p.m., the price and storage level are fluctuating and the storage level reaches a peak at 4 p.m. During 5 p.m. and 10 p.m., the electricity price is relatively high and users start to discharge. At the end of this day, the SES storage level returns to the initial state, completing a daily cycle.

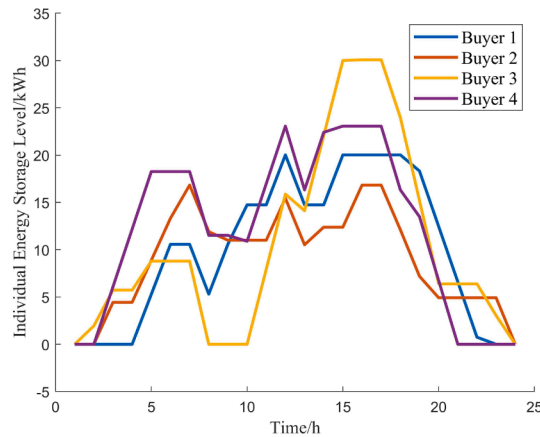
5.2.4. Shared energy storage versus individual energy storage

To validate the potential for sharing the energy storage, the operational costs of individual and shared energy storage are compared, which are the results of Case 3 and Case 4, respectively. To ensure a fair comparison, the default parameter settings in the individual scenario are consistent with the SES scenario. As for the capacity allocation, we

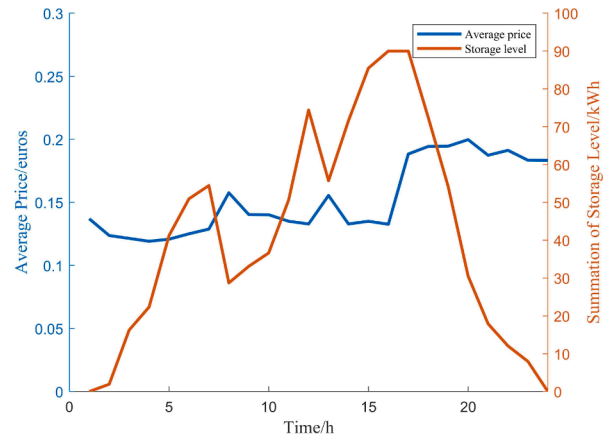
assume the capacity is installed proportional to the daily energy demand level, which means more energy demand accompanied by larger capacity [49]. The solution method is the same but much easier than the SES scenario because of the lack of shared constraint (25) in the individual scenario model.

The operational cost comparison is summarized in Table 6, which comprises personal energy cost, personal capacity allocation, and the total cost of all buyers. Compared with no energy storage device, buyers installing individual energy storage can receive a lower energy cost. The total cost savings is 4.40%. The result is satisfactory and reasonable because buyers can manage their energy consumption more flexibly and achieve peak-valley arbitrage by utilizing individual storage devices. Besides, it can be seen that, compared with individual storage, shared energy storage can gain more energy cost saving, up to 5.46%. The comparison indicates that shared energy storage has larger economic potential than individual storage. This is because, in our model, the SES capacity is allocated to those who really need energy storage and can receive a larger cost reduction by using the same storage capacity.

Fig. 6a shows the one-day individual energy storage level for each buyer. Fig. 6b shows the operation of individual energy storage devices from a holistic perspective. The left coordinate axis corresponds to the average energy price for 4 buyers, with the same meaning in Fig. 5b. The right coordinate axis corresponds to the summation of individual energy storage levels. During 1 a.m. and 7 a.m., the electricity price is relatively low and users start to charge. During 8 a.m. and 4 p.m., the price and storage level are fluctuating and the storage level reaches a peak at 4 p.m. During 5 p.m. and 12 p.m., the electricity price is relatively high and users start to discharge individual storage. At the end of this day, the individual storage level returns to the initial state, completing a daily cycle.



(a) Individual energy storage level for 4 buyers



(b) Summation of individual energy storage level and average price

Fig. 6. One day operation of individual energy storage.

6. Conclusion

Peer-to-peer energy trading is believed to be a promising business model in the future transactive energy market, which enables DERs owners to share excessive energy with local peers. P2P energy trading can bring financial benefits, balance local energy supply and demand, and improve energy security. The stakeholder's structure and energy pricing mechanism are two points of concern that should be taken into account when designing a P2P energy trading mechanism. In this paper, we have designed a P2P energy trading mechanism for local DERs owners and residential consumers embedded with residential shared energy storage, in which each peer aims to maximize personal interest. The implementation of shared energy storage can provide energy buyers more flexible energy consumption strategy. The first game between energy sellers and buyers and the second game among energy buyers make the problem an equilibrium model. An aggregator is assigned to operate the equilibrium model. Based on the Karush-Kuhn-Tucker optimality conditions of all participants' optimization problems and linearization techniques, the equilibrium condition can be cast as a MILP, in which the P2P transaction price and capacity allocation are endogenously determined. Numerical results demonstrate the effectiveness of the proposed P2P market, both energy sellers and buyers benefiting from P2P energy trading. Besides, P2P energy trading facilitates the consumption of local renewable energy. The implementation of shared energy storage can further reduce energy costs and improve local renewable consumption.

In future work, the uncertainty of renewable energy generation and load profile has to be considered. Besides, the sellers should also utilize energy storage to flexibly manage their supply plan to earn more profit. We just focus on the operation of shared energy storage in this paper, the investment decision, such as capacity sizing, should be evaluated for a long period.

CRedit authorship contribution statement

Boshen Zheng: Conceptualization, Software, Writing – original draft. **Wei Wei:** Methodology, Data curation, Writing – review & editing. **Yue Chen:** Visualization, Investigation, Writing – review & editing. **Qiuwei Wu:** Supervision, Validation, Writing – review & editing. **Shengwei Mei:** Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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